Tenext Voice AI: SIP Interconnect

Overview

Tenext.ai is a Philippines-based Voice AI company building intelligent voicebots integrated into existing contact center infrastructure. One of our key partners is **BDO**, a major financial institution with an **on-premise Avaya Aura** setup that handles inbound customer calls over local PSTN lines.

Tenext's voice bots run on Daily's Pipecat Cloud, the bots are hosted in **AWS US-West**. The goal is to interconnect BDO's on-premise Avaya infrastructure with Daily.co's infrastructure. At peak BDO is expected to do 600 simultaneous calls which would be about 100 Mbps.

Architecture Summary

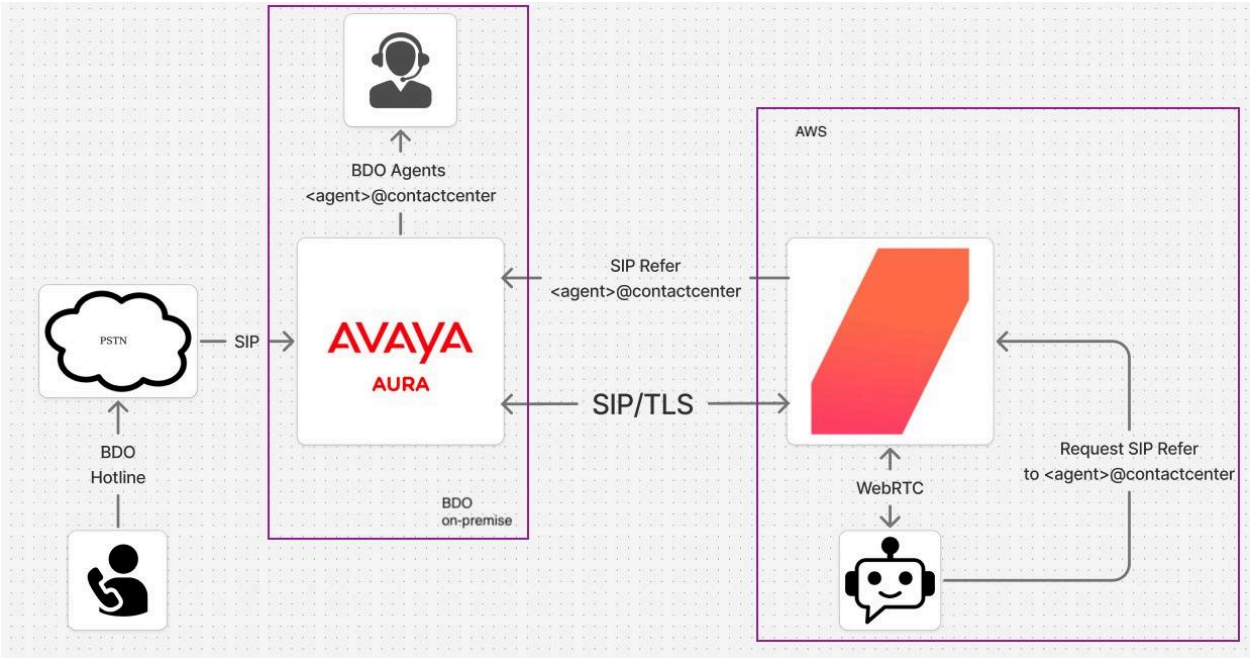
- **BDO** hosts on-premises **Avaya Aura Server**, which handles PSTN calls locally in the Philippines.

- **Avaya** connects to **Daily.co SIP infrastructure** via SIP (UDP/TCP/TLS/WebSockets) and RTP (encrypted via DTLS).

- **Daily** proxies SIP signaling and RTP media to Daily's **Pipecat cloud**.

Technical and Bandwidth Requirements

- **Protocol support**: SIP over UDP/TCP/TLS/WebSockets; RTP over DTLS (SRTP)
- **Codec support**: G.711 (PCMU),
- **Encryption**: DTLS by default;
- **Max expected Load**: ~600 simultaneous calls
- **Bandwidth estimate**: ~100 Mbps total RTP traffic at peak (based on G.711 @ 20ms packetization)



The Avaya server is not currently connected to the Internet and thus BDO is exploring several options. In either of the proposed case, they will have to standup infrastructure at their end to connect to Daily's infrastructure. Either it is a VPN tunnel that

1. VPN Tunnel / Leased Line is a direct tunnel from BDO's secure network to Daily's AWS infrastructure. The tunnel provides a stronger control over security and routing, however, can be a signle point of failure, especially, ongoing calls will immediately disconnect if the tunnel fails.

2. Public Internet + IP Whitelisting: BDO whitelists Daily's published IP ranges. SIP/RTP media sent over the public Internet to Daily POPs in Asia (e.g., Singapore, South Korea). We have more flexibility for scaling and reliability

From AWS for option 1 can be implemented using AWS Site-to-Site VPN.

- https://docs.aws.amazon.com/vpn/latest/s2svpn/Examples.html
- https://docs.aws.amazon.com/vpn/latest/s2svpn/vpn-redundant-connection.html
- Each AWS Site-to-Site VPN connection has two tunnels and each tunnel supports a maximum throughput of up to 1.25 Gbps